

Dr. Rajesh Nayak

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OBJECTIVE

To contribute effectively in teaching, research, and academic development by utilizing my knowledge, skills, and experience in the field of Computer Science and Engineering. I aim to work in a dynamic academic environment that encourages learning, innovation, and professional growth while making meaningful contributions to students and the institution.

WORK EXPERIENCE

Associate Professor & HoD January 2026 – Present
Dept. of AI & ML, Shri Madhwa Vadiraja Institute of Technology and Management, Bantakal, Udupi, Karnataka, India.

Associate Professor September 2024 – January 2026
Dept. of AI & DS, Shri Madhwa Vadiraja Institute of Technology and Management, Bantakal, Udupi, Karnataka, India.

Research Scholar and Teaching Assistant October 2020 – September 2024
Dept. of Information Technology, National Institute of Technology Karnataka, Surathkal, Mangaluru, Karnataka, India.

Assistant Professor May 2019 – September 2020
Dept. of CS & E, Sahyadri College of Engineering and Management, Mangaluru, Karnataka, India.

Assistant Professor June 2013 – November 2018
Dept. of CS & E, Moodlakatte Institute of Technology, Kundapura, Udupi, Karnataka, India.

EDUCATION

2020 – 2024	PhD (Data Analytics in Smart Grid) at Dept. of Information Technology, NITK Surathkal (Thesis submitted on 26th November 2024. Defended on 20th May 2025. Convocation on 20th November 2025)	
2011 – 2013	M.Tech. (CSE) at JNNCE Shivamogga	(Percentage: 77)
2007 – 2011	B.E. (ISE) at CIT Gubbi, Tumkur	(Percentage: 69)

PROJECTS

- Smart Grid Analytics** Developed projects on electricity forecasting, electricity theft detection, and anomalous power consumption detection using Machine Learning and Deep Learning techniques for intelligent energy management.
- Smart Plug-based Power Monitoring System** Designed and implemented anomaly detection models using ML/DL by integrating Smart Plug hardware with software systems for real-time monitoring and analysis of household power consumption.
- Healthcare Analytics** Worked on Explainable AI-based medical imaging projects such as bone fracture detection using deep learning techniques for improved diagnosis and interpretation.

- **Precision Agriculture** Developed plant disease detection systems such as Jasmine plant disease identification using image processing, multi-data fusion, and deep learning approaches.

Current Research and Development Projects

- Pepper plant disease detection and harvest readiness prediction using Artificial Intelligence techniques.
- Development of a Smart Plug system for multiple device connectivity and intelligent power monitoring in the presence and absence of consumers.
- Smart navigation stick for visually impaired individuals using intelligent sensing and navigation technologies.
- Women safety system using a multi-modal approach integrating intelligent monitoring and alert mechanisms.

RESEARCH INTERESTS

1. Smart Grid Analytics: Electricity Theft Detection, Power Forecasting, Anomaly Detection using ML/DL.
2. Healthcare Analytics: Explainable AI in Medical Imaging (e.g., bone fracture detection). Disease Diagnosis using ML/DL.
3. Precision Agriculture: Explainable AI in Plant Disease Diagnosis, Plant Health Monitoring.
4. Cross-domain applications of Machine Learning and Deep Learning.

PUBLICATIONS ([GOOGLE SCHOLAR](#) AND [ORCID](#))

International Journals

1. **Rajesh Nayak and Jaidhar C D**, “Anomalous Electrical Power Consumption Detection in Household Appliances via Micro-Moment Classification,” IEEE Transaction on Artificial Intelligence, IEEE. [**Scopus and SCIE (Q1)**] <https://doi.org/10.1109/TAI.2025.3622104>.
2. **Rajesh Nayak and Jaidhar C D**, “Enhancing Anomaly Detection in Critical Systems Using Household Appliance Power Consumption Data,” IEEE Sensors Journal, 24, 17, 27677-27686 (2024). [**Scopus and SCIE (Q1)**] <https://doi.org/10.1109/JSEN.2024.3426090>.
3. **Rajesh Nayak and Jaidhar C D**, “Data-driven models for electricity theft and anomalous power consumption detection: a systematic review”, Applied Intelligence 55, 794 (2025). [**Scopus and SCIE (Q2)**] <https://doi.org/10.1007/s10489-025-06663-3>
4. **Rajesh Nayak and Jaidhar C D**, “Employing Feature Extraction, Feature Selection, and Machine Learning to Classify Electricity Consumption as Normal or Electricity Theft”, Recent Trends in Computational Intelligence, SN Computer Science, 4, 483 (2023). [**Scopus (Q2)**] <https://doi.org/10.1007/s42979-023-01911-0>.

International Conferences

1. **Rajesh Nayak and Jaidhar C D**, “Micro-Moment Classification for Anomalous Power Consumption Detection using 1D CNN.” In 2023 Third International Conference on Secure Cyber Computing and Communication (ICSCCC), IEEE, 97-102 (2023). <https://doi.org/10.1109/ICSCCC58608.2023.10176687>
2. **Rajesh Nayak and Jaidhar C D**, “Classification of Micro-Moment-Based Anomalous Power Consumption Using Transfer Learning.” In 2023 5th International Conference on Energy, Power

and Environment: Towards Flexible Green Energy Technologies (ICEPE), IEEE, 1-6 (2023).
<https://doi.org/10.1109/ICEPE57949.2023.10201605>

3. **Rajesh Nayak and Jaidhar C D**, “Experimental Study on Detection of Household Electrical Appliance Energy Consumption Deviation,” accepted and presented in 3rd International Symposium on Sustainable Energy and Technological Advancements (ISSETA 2024), held at NIT Meghalaya, India, February 23-24, 2024.
4. **Rajesh Nayak and Jaidhar C D**, “QR Code-based Anomalous Power Consumption Detection,” accepted and presented in International Conference on Industrial Engineering and Analytics (ICONIEA 2024), held at IIT Kharagpur, February 16-18, 2024.
5. **Rajesh Nayak and Jaidhar C D**, “Smart Appliance Abnormal Electrical Power Consumption Detection.” In: Andreoni, M. (eds) Applied Cryptography and Network Security Workshops. ACNS 2024. Lecture Notes in Computer Science, vol 14587. Springer, Cham.
https://doi.org/10.1007/978-3-031-61489-7_13 (Core B).
6. **Rajesh Nayak and Jaidhar C D**, “Experimental Study on Impact of Appliance ID-Based Normalization on SimDataset for Anomalous Power Consumption Classification,” 17th International Conference on Security of Information and Networks (SIN’24), 1-8, held at University of Technology Sydney, 15 Broadway Ultimo, NSW Australia, December 02-04, 2024.
<https://doi.org/10.1109/SIN63213.2024.10871753>. (Core C).
7. **Rajesh Nayak and Jaidhar C D**, “POSTER: Electricity Theft Detection using Machine Learning,” accepted and presented in 18th International Conference on Information and Systems Security (ICISS) 2022, held at IIT Tirupati, December 16-20, 2022.
8. **Rajesh Nayak., et. al**, “Affordable Camera-Based Assistive Navigation System for Visually Impaired Individuals.” In International Conference on Materials, Energy, Environment & Manufacturing Sciences & Computational Intelligence and Smart Communication (MEEMS-CISC-2024) (pp. 159-167). Atlantis Press. https://doi.org/10.2991/978-94-6463-762-5_15.
9. **Rajesh Nayak., et. al**, “State-level Power Consumption Forecasting using Machine Learning.” 4th International Conference on Data, Decision and Systems (ICDDS’25), to be held at the Indian Institute of Information Technology, Dharwad, India, from December 4-6, 2025 (IEEE).(Accepted).
10. **Rajesh Nayak., et. al**, “AI-driven Dried Coconut Classification Using Computer Vision.” Second International Conference on Computing, Semiconductor, Mechatronics, Intelligent Systems and Communications (COSMIC - 2025), to be held at Sahyadri College of Engineering and Management, Mangalore, November 21-22, 2025 (IEEE).(Accepted).
11. **Rajesh Nayak., et. al**, “Machine Learning Based Forecasting of State-Level Electricity Demand.” International Conference on Advance Smart, Secure and Intelligent Computing (ASSIC’26), to be held at KIIT-DU Campus, Bhubaneswar, India, from August 8-10, 2026 (IEEE).(Accepted).

SKILLS

Technical Skills: Programming Languages (C, Basics of Java, Python, SQL), Database Management, Data Analytics, Machine Learning, Deep Learning, and AI Knowledge. Hadoop, Big Data Analytics on Cloudera, Big Data Analytics framework like Hive and Pig

Soft Skills: Communication (Written and Verbal), Teamwork and Collaboration, Problem-solving, Time management, Critical thinking, and Leadership.

SUBJECTS HANDLED (MAJOR)

- Deep Learning and Reinforcement Learning with Lab, Big Data Analytics with Lab, Software Engineering and Project Management. Database Management System with lab and Mini project, Information & Network Security, Computer Networks, Design & Analysis of Algorithms.

RESPONSIBILITIES HANDLED

- Academic and administrative responsibilities including Program Coordinator, Academic Coordinator, Class Coordinator, and Placement Coordinator.
- Coordinated projects, seminars, workshops, magazine activities, and TEDx events at the institute level.
- Served as Institute Webmaster and contributed as an NBA Core Committee Member and Coordinator.
- Worked as Faculty Coordinator for DDU-GKY (Deen Dayal Upadhyaya Grameen Kaushalya Yojana), a Government of India initiative.

PROFESSIONAL ACTIVITIES & ACHIEVEMENTS

Academic Contributions

- Resource Person for 2 days hands-on workshop on IoT, Deep Learning and its Applications at AJIET Mangaluru. Resource Person for a session on Intellectual Property Rights at CIT Gubbi, Tumakuru. Resource Person for AICTE-QIP-PG session on *Sequential Modeling* at NITK Surathkal.
- Served as Session Chair and Session Co-Chair at CISCON 2025, MIT Manipal.
- Reviewer for multiple Scopus-indexed international conference articles.

Grants & Recognitions

- Received \$600 student travel grant to attend ACNS-2024 at NYU Abu Dhabi.
- Received travel grant for ICISS-2022 at IIT Tirupati.
- Awarded MHRD stipend-based full-time Ph.D. admission at NITK Surathkal.
- Received KSCST funding support for three final year B.E. projects.
- Qualified UGC-NET examination in 2018.

Faculty Development Programs & Workshops

- Attended AICTE ATAL FDP on Smart Agriculture and FDP on Quantum Computing.
- Participated in FDPs/workshops on OBE, MATLAB, Cloud Computing, and Open Source Technologies.

DECLARATION

I hereby declare that all the information given above is true and correct to the best of my knowledge.